Power analysis on 8Ã— MI300X

openai/gpt-oss-120b

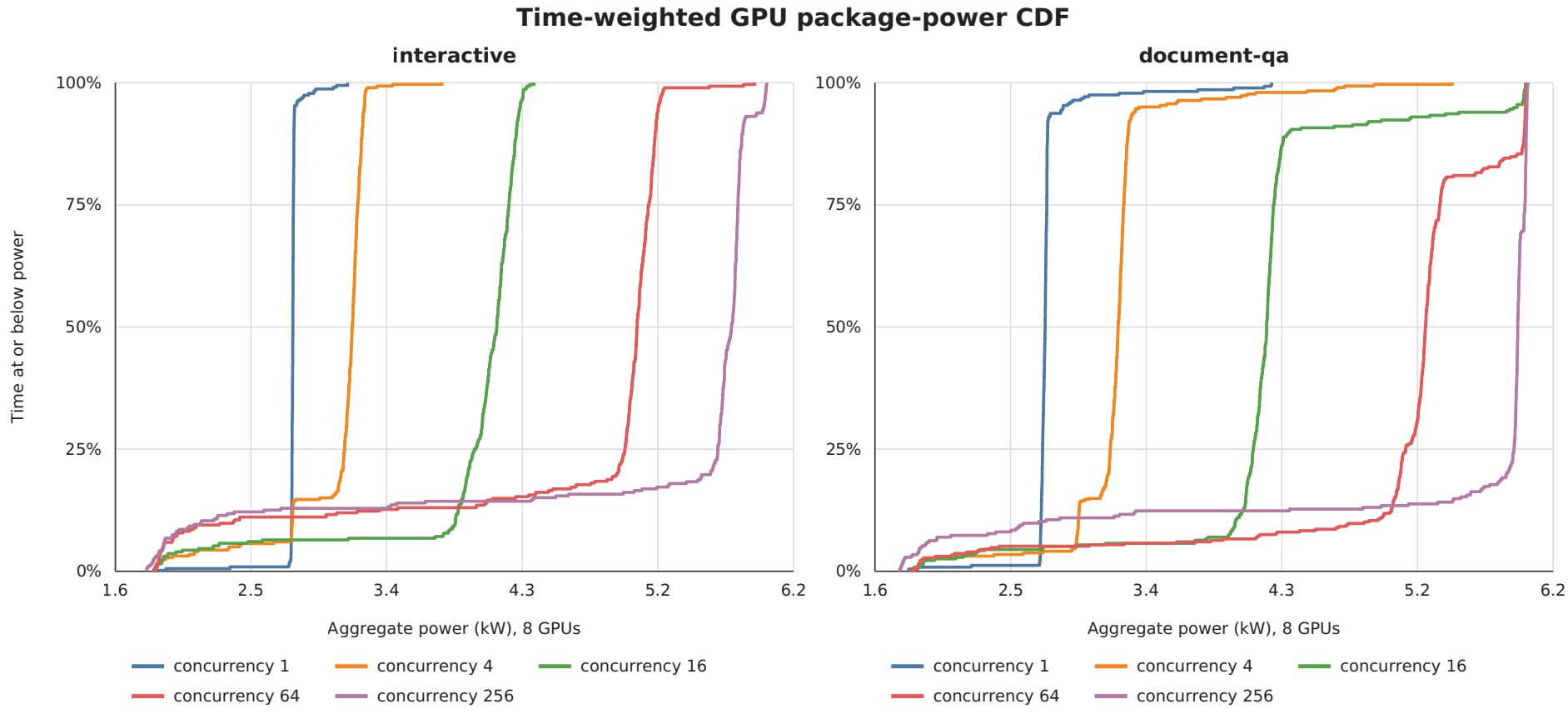
All 24 measurements completed without failed requests. Power is the sum of the eight GPU package readings, not whole-node wall power. `%TDP` uses AMD SMI's reported 750 W maximum package power per GPU (6.0 kW total) as a proxy.

Overview

Run	ISL/OSL	Conc	Time	Out tok/s	Avg W	Max sampled W	TDP %
decode-heavy-i8192-o8192-c1	8192/8192	1	221.16s	148.2	2,759.9	2,778.0	46.0%
decode-heavy-i8192-o8192-c4	8192/8192	4	146.65s	446.9	3,200.2	3,332.0	53.3%
decode-heavy-i8192-o8192-c16	8192/8192	16	160.23s	1,636.1	4,149.9	4,394.0	69.2%
decode-heavy-i8192-o8192-c32	8192/8192	32	176.51s	2,970.3	4,708.6	4,988.0	78.5%
document-qa-i8192-o1024-c1	8192/1024	1	70.07s	146.1	2,746.2	4,258.0	45.8%
document-qa-i8192-o1024-c4	8192/1024	4	75.49s	406.9	3,196.0	5,482.0	53.3%
document-qa-i8192-o1024-c16	8192/1024	16	78.61s	1,432.9	4,222.5	5,981.0	70.4%
document-qa-i8192-o1024-c64	8192/1024	64	84.25s	3,682.7	5,177.0	5,988.0	86.3%
document-qa-i8192-o1024-c256	8192/1024	256	69.98s	7,491.9	5,450.0	5,995.0	90.8%
interactive-i1024-o1024-c1	1024/1024	1	68.89s	148.6	2,771.2	3,143.0	46.2%
interactive-i1024-o1024-c4	1024/1024	4	74.87s	396.6	3,083.4	3,784.0	51.4%
interactive-i1024-o1024-c16	1024/1024	16	72.20s	1,574.2	4,003.0	4,407.0	66.7%
interactive-i1024-o1024-c64	1024/1024	64	72.02s	4,293.8	4,711.6	5,903.0	78.5%
interactive-i1024-o1024-c256	1024/1024	256	68.89s	11,013.8	5,210.3	5,984.0	86.8%
large-kv-32k-i32768-o2048-c1	32768/2048	1	72.62s	141.0	2,773.0	5,808.0	46.2%
large-kv-32k-i32768-o2048-c4	32768/2048	4	79.98s	384.1	3,360.0	5,955.0	56.0%
large-kv-32k-i32768-o2048-c8	32768/2048	8	80.75s	710.1	3,829.0	5,952.0	63.8%
large-kv-32k-i32768-o2048-c16	32768/2048	16	91.06s	1,147.1	4,231.5	5,970.0	70.5%
large-kv-64k-i65536-o4096-c1	65536/4096	1	116.08s	141.1	2,879.6	5,953.0	48.0%
large-kv-64k-i65536-o4096-c4	65536/4096	4	80.92s	404.9	3,461.7	5,975.0	57.7%
large-kv-64k-i65536-o4096-c8	65536/4096	8	86.35s	758.9	4,033.0	5,973.0	67.2%

Run	ISL/OSL	Conc	Time	Out tok/s	Avg W	Max sampled W	TDP %
large-kv-64k-i65536-o4096-c16	65536/4096	16	103.97s	1,260.6	4,665.9	5,986.0	77.8%
smoke-i256-o128-c1	256/128	1	43.69s	146.5	2,790.9	2,982.0	46.5%
smoke-i256-o128-c4	256/128	4	44.82s	419.8	3,197.0	3,824.0	53.3%

Time-weighted power CDF



For any power value on the X-axis, the Y-axis is the percentage of benchmark time spent at or below it. A curve farther right means more sustained high power; a brief maximum has little effect on most of the curve.

Power conclusions

- **Batching improves power utilization.** Interactive c64â†’c256 raises throughput 156.5% while average power rises 10.6%. Document-QA throughput rises 103.4% for 5.3% more power. More concurrent tokens amortize model-weight traffic and keep compute, HBM, and the tensor-parallel fabric busier.
- **High concurrency sustains near-limit power.** At c256, median power is 5.75 kW for interactive and 5.93 kW for document-QA; their 90th percentiles are 5.82 and 5.98 kW. The right-shifted CDF curves show this is sustained, not just a momentary spike.
- **Maximum power alone is misleading.** Large-KV c1 briefly reaches 5.81â€“5.95 kW during prompt processing, but averages only 2.77â€“2.88 kW. Its long decode phase runs much lower, so the brief prefill peak barely affects average power.
- **Decode-heavy c32 is genuinely high-power decode.** Its middle 80% of power samples lies around 4.60â€–4.86 kW, while median TTFT is only 0.67 s within a 176.5 s run. Prefill/decode overlap is therefore not the main cause: batching 32 decode tokens turns small operations into more efficient matrix work and raises sustained package utilization.
- **Low concurrency leaves substantial headroom.** Across short, interactive, document, and decode-heavy workloads, c1 averages about 2.75â€–2.88 kW, or 46â€–48% of the 6.0 kW proxy TDP. The model is resident on all eight GPUs, but too little work is in flight to use the available power envelope.

Combined text analysis Â· 22-point baseline Â· c256 follow-up

MiniMaxAI/MiniMax-M3-MXFP8

Interim snapshot: 11 of 24 planned measurements are complete. The live suite is continuing with `document-qa-i8192-o1024-c256`; incomplete points are not failures and are omitted from this table. Power is the sum of the eight GPU package readings. `% TDP` uses AMD SMI's reported 750 W maximum package power per GPU (6.0 kW total) as a proxy.

Overview

Run	ISL/OSL	Conc	Time	Out tok/s	Avg W	Max sampled W	TDP %
document-qa-i8192-o1024-c1	8192/1024	1	79.24s	90.5	3,556.4	5,852.0	59.3%
document-qa-i8192-o1024-c4	8192/1024	4	80.09s	294.1	4,772.5	5,959.0	79.5%
document-qa-i8192-o1024-c16	8192/1024	16	87.92s	652.2	5,629.6	5,973.0	93.8%
document-qa-i8192-o1024-c64	8192/1024	64	119.53s	1,096.6	5,959.9	6,008.0	99.3%
interactive-i1024-o1024-c1	1024/1024	1	76.44s	93.8	3,543.8	4,721.0	59.1%
interactive-i1024-o1024-c4	1024/1024	4	73.20s	321.8	4,720.5	5,558.0	78.7%
interactive-i1024-o1024-c16	1024/1024	16	74.38s	812.3	5,718.6	5,981.0	95.3%
interactive-i1024-o1024-c64	1024/1024	64	86.06s	1,523.0	5,963.2	6,009.0	99.4%
interactive-i1024-o1024-c256	1024/1024	256	139.60s	3,755.8	5,971.1	6,007.0	99.5%
smoke-i256-o128-c1	256/128	1	70.34s	89.2	3,625.6	4,028.0	60.4%
smoke-i256-o128-c4	256/128	4	57.07s	298.3	4,611.6	5,188.0	76.9%